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SECTION: A

USE CASE FLOW SPECIFICATION

Use-Case Flow Specification

System: Local Sports Tournament & Bracket Manager

Use Case ID: UC-02

Use Case Name: Generate Tournament Bracket

Primary Actor: Tournament Director

Related Use Case: UC-08 – Validate Team Rankings

Preconditions

1. The Tournament Director is authenticated and authorized to manage the tournament.
2. Teams have been registered in the system.
3. Each registered team has a valid initial ranking.
4. A valid tournament format, either single-elimination or double-elimination, has been selected.

Postconditions

1. The tournament bracket is successfully generated.
2. All participating teams are assigned to valid bracket positions according to their rankings.
3. The generated bracket is stored and made available for viewing.
4. The tournament is ready for match progression.

Main Success Scenario

1. Tournament Director selects "Generate Tournament Bracket."
2. System displays the registered teams, their initial rankings, and the available tournament formats.
3. Tournament Director selects the required tournament format: single-elimination or double-elimination.
4. System validates the team rankings using UC-08 – Validate Team Rankings.
5. System determines the seeded positions of the participating teams based on their initial rankings.
6. System generates the tournament bracket and assigns each team to a valid bracket slot.
7. System displays the generated bracket to the Tournament Director.
8. Tournament Director reviews and confirms the generated bracket.
9. System saves the confirmed bracket and makes it available for tournament use.
10. Use case ends successfully.

Alternate Flow

A1. Invalid or Missing Team Rankings

4a. System detects that one or more teams have missing, duplicate, or invalid rankings.

4a1. System displays an error message identifying the invalid or missing ranking information.

4a2. System does not generate the tournament bracket.

4a3. Tournament Director corrects the team ranking information.

4a4. System revalidates the rankings.

4a5. If all rankings are valid, the system resumes the Main Success Scenario at Step 5.

4a6. If the rankings remain invalid, the bracket-generation process is cancelled.

End of Use Case